MILITARY SPECIFICATION

MOTOR, HOT GAS

(MAU-132A/B)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the MAU-132A/B Hot Gas Motor, critical item, referred to herein as the motor.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-45662	Calibration System Requirements.
MIL-B-85251	Bomb, Binary Chemical, BLU-80/B.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification and Marking of U.S. Military Property.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-45662	Calibration System Requirements.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL X4900750	Motor, Hot Gas MAU-132A/B.
SA 2875556	Fixture, Vibration (MAU-132A/B, BBU-18/B).
SA 495191	Orifice Gage Adapter, Static Firing, Gas Generator.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations

49 CFR 171-178 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

3.1 Item description. The motor is a device powered by a gas pressure generator (BBU-18/B). The major components of the motor are a cylinder block, wobble plate assembly, inlet valve assembly, pistons, and motor mount.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Functional (cold gas). When assembled and operated on nitrogen gas, the motor performance shall meet the requirements of Figures 1 and 2 when loaded to extract 7 horsepower at 2,200 ±100 revolutions per minute (rpm).

3.2.2 Environmental. The motor performance shall not be degraded below that specified in Figures 1 and 2 after exposure to the following environments.

3.2.2.1 High temperature. The motor shall withstand exposure to a high temperature of 74 degrees Celsius (°C) for 48 consecutive hours.

3.2.2.2 Low temperature. The motor shall withstand exposure to low temperature of -48°C for 24 consecutive hours.

3.2.2.3 Vibration. The motor shall withstand exposure to the vibration schedule specified in Figure 3 in each of three mutually perpendicular axes in accordance with 4.3.2.3.

3.2.2.4 Mechanical shock. The motor shall withstand exposure to half sine wave impact shocks in each of three mutually perpendicular axes having a peak amplitude of 15 gravity units (g) and a total duration of 11 milliseconds (ms) at a temperature of -48°C.

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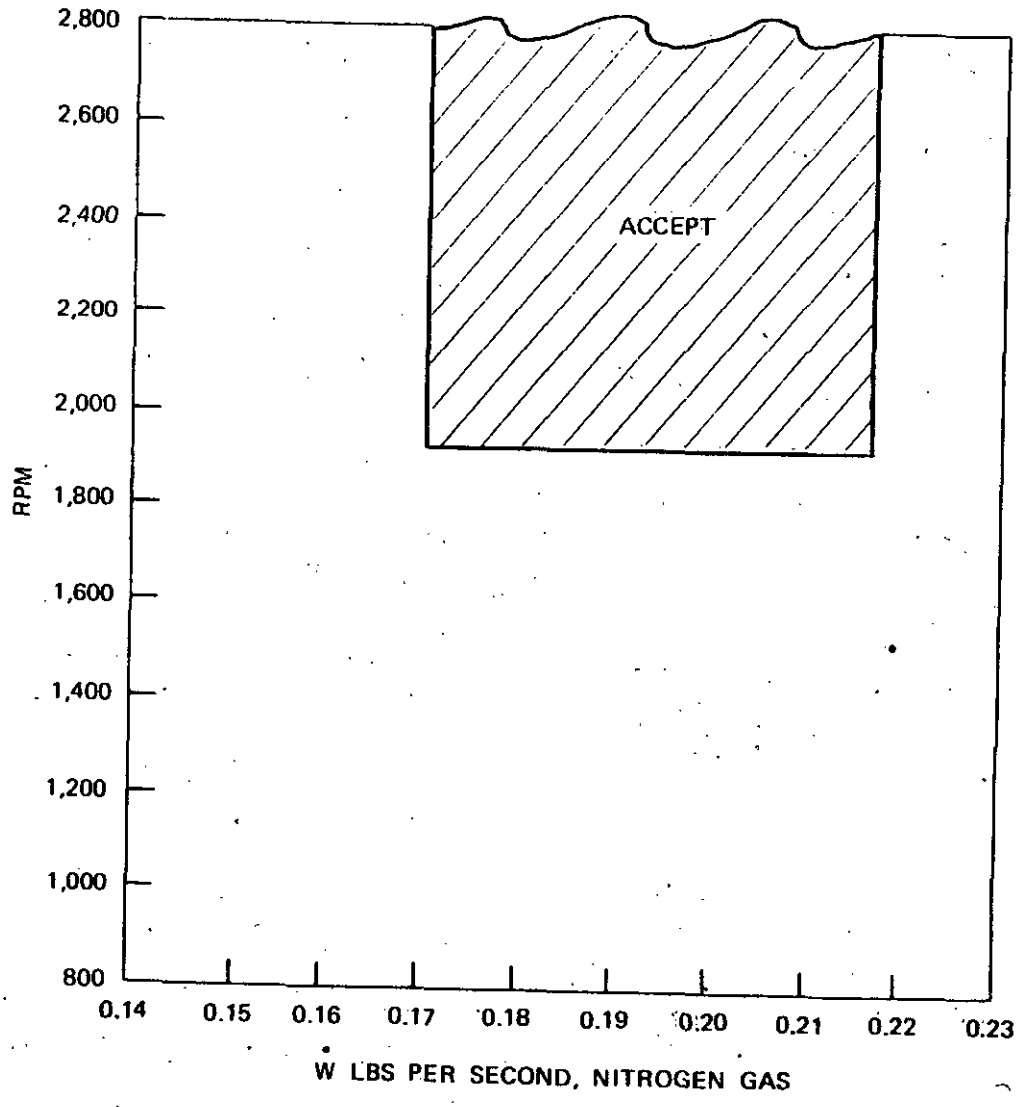


FIGURE 1. RPM performance envelope.

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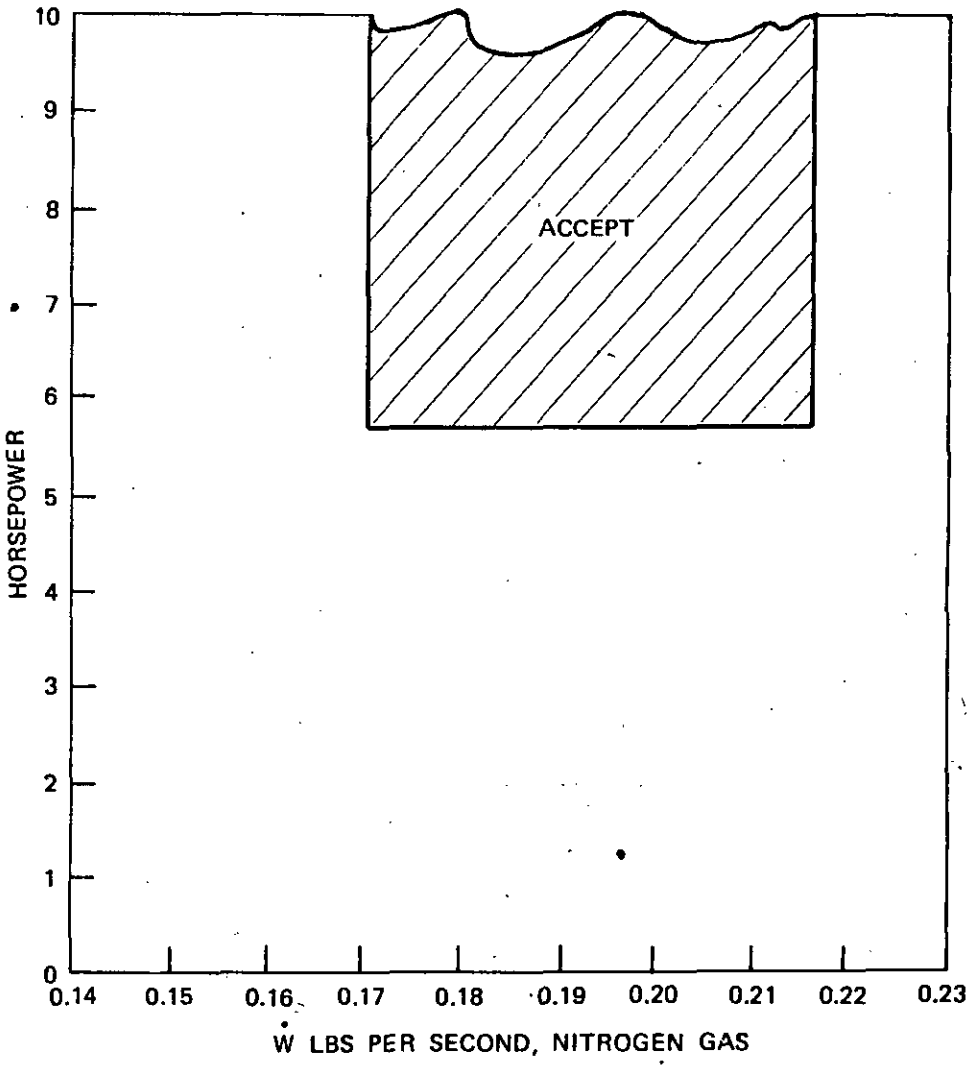


FIGURE 2. Horsepower performance envelope.

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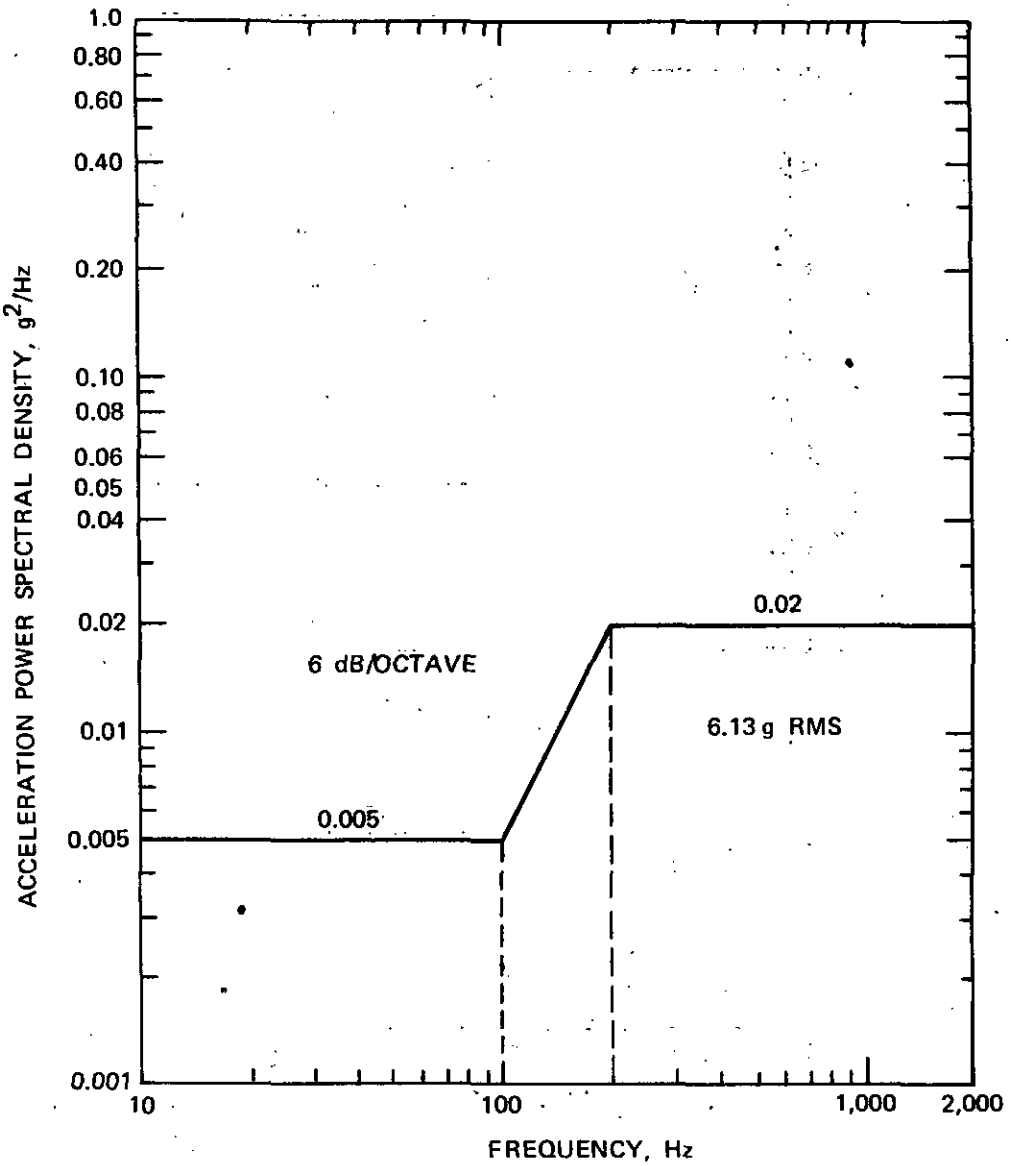


FIGURE 3. Vibration schedule.

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3.2.2.5 Temperature shock. The motor shall withstand exposure to sequential alternating temperatures of -48°C and +74°C.

3.3 Design and construction.

3.3.1 Production drawings. The motor shall be fabricated and assembled in accordance with the drawings, parts list, and other documents listed on DL X4900750, and this specification.

3.3.2 Standards of manufacture.

3.3.2.1 Working environment. Special working environments for the manufacture of the motor shall be as specified in MIL-B-85251.

3.3.2.2 Certification. When specified in the contract or purchase order (see 6.2.2), the contractor shall include written certification accompanied by objective quality evidence as defined in MIL-STD-109, that the materials, processes, and parts used in the motor meet the requirements of Section 3.

3.3.3 Marking. Each motor shall be marked in accordance with MIL-STD-130, with the manufacturer's name or symbol, part number and manufacturer's lot designation. The markings shall be as shown on DL X4900750.

3.3.4 Cold gas spin-up. When operated on nitrogen gas such that motor speed is 2,000 rpm minimum, the motor shall operate smoothly without binding.

3.4 First article. Unless otherwise specified in the contract or purchase order, the contractor shall furnish a preproduction sample consisting of 15 motors to the testing activity designated in the contract or purchase order for first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is totally at the contractor's risk. Any changes in design materials or processes used in the manufacture of the preproduction sample may be cause for resubmission of another preproduction sample for testing.

3.4.1 Retest. At the discretion of the procuring activity, preproduction tests, or any portion thereof, shall be repeated under any of the following conditions:

a. The manufacturer has modified his product (such as a change of raw materials, the process, production procedures, or methods).

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b. Where there is evidence that the quality of the product has not been maintained. The evidence may be in the form of accumulated product failures, of system failures attributable to the product, or failure of the product to pass any of the tests for production lot acceptance that may be conducted by or for the procuring activity.

c. When applicable documents have been amended or revised sufficiently so that continued validity of the previous preproduction testing is questionable.

d. When the manufacturer has not produced motors to this specification for a period of 12 months or longer.

e. When the manufacturer has changed the location at which the motors are produced.

3.5 Documentation. When specified in the contract or purchase order (see 6.2.2), inspection reports shall be prepared in accordance with 4.1.2.3 and 4.2.3.

3.6 Workmanship. As a minimum, the motor shall be manufactured, inspected, and tested under conditions and procedures meeting the requirements of 3.3.1 and as specified herein. The motor shall be fabricated in a manner that will ensure compliance with all requirements of this specification and the assembly drawing. Unless otherwise specified herein or on the assembly drawing, fabrication and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Classification of inspections. The inspection of the motor shall be classified as follows:

a. Preproduction inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction inspection. The preproduction sample of 3.4 shall be subjected to the examinations and tests of Table I in the sequence shown. Group I quality conformance inspections (see Table II) shall be performed on all preproduction motors offered for acceptance.

TABLE I. Preproduction inspections.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical	3.3.1, 3.6	4.3.1
Functional (cold gas)	3.2.1.1	4.3.3
High temperature	3.2.2.1	4.3.2.1
Functional (cold gas)	3.2.1.1, 3.2.2	4.3.3
Low temperature	3.2.2.2	4.3.2.2
Functional (cold gas)	3.2.1.1, 3.2.2	4.3.3
Vibration	3.2.2.3	4.3.2.3
Functional (cold gas)	3.2.1.1, 3.2.2	4.3.3
Temperature shock	3.2.2.5	4.3.2.5
Functional (cold gas)	3.2.1.1, 3.2.2	4.3.3
Mechanical shock	3.2.2.4	4.3.2.4
Functional (cold gas)	3.2.1.1, 3.2.2	4.3.3

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TABLE II. Quality conformance inspections.

Examination or test	Requirement paragraph	Method paragraph
Group I (100%)		
Visual and mechanical	3.3.1, 3.6	4.3.1
Cold gas spin-up	3.3.4	4.3.4
Group II (sample)		
Functional	3.2.1.1	4.3.3

4.1.2.2 Acceptance criteria. Failure of one or more preproduction samples to pass any of the preproduction tests shall cause rejection of all preproduction samples.

4.1.2.3 Preproduction inspection report. When specified in the contract or purchase order (see 6.2.2), results of the preproduction inspections shall be prepared for the procuring activity. The report shall contain the following information accompanied by a certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the preproduction sample complies with all quality assurance provisions of this specification for the motor specified in the contract or purchase order.

b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Contract or purchase order number and date, together with an identification and date of changes.

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e. Certificates of compliance of all material procured directly by the contractor.

f. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Quality conformance inspections. Quality conformance inspections shall be performed on each lot of motors as specified in Table II, in the sequence shown.

4.2.1 Sampling. Each motor shall be visually inspected and subjected to a cold gas spin-up prior to its incorporation into a lot. A random sample shall be drawn from each lot for Group II inspections of Table II. The sample size shall be in accordance with MIL-STD-105, Inspection Level II, Normal Inspection, Single Sampling, and an acceptable quality level (AQL) of 1.0 percent defective.

4.2.2 Acceptance criteria. Failure of one or more motors in the quality conformance sample to pass the Group II tests shall cause rejection of the lot represented.

4.2.3 Quality conformance inspection report. When specified in the contract or purchase order (see 6.2.2), results of the quality conformance inspection shall be included with each lot. Reports shall contain the following information accompanied by a certification (see 6.2.2) which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that each motor complies with all quality assurance provisions of this specification for the motor specified in the contract or purchase order.

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b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Contract or purchase order number and date together with an identification and date of changes.

e. Certificates of compliance of all material procured directly by the contractor.

f. Date submitted.

The certification shall be signed by a responsible agent of the certifying organization. The initial certification shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certifications unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3 Inspection methods.

4.3.1 Visual and mechanical examination. Each motor shall be inspected for conformance to the requirements of 3.3.1 and 3.6.

4.3.2 Environmental tests.

4.3.2.1 High temperature. The motor samples shall be subjected to the high temperature test of MIL-STD-810, Method 501.1, Procedure I, except the temperature shall be 74°C. Steps 4 and 5 shall be omitted. The motor samples shall then be operated in accordance with 4.3.3. To be acceptable, the motors shall meet the requirements of 3.2.1.1.

4.3.2.2 Low temperature. The motor samples shall be subjected to the low temperature test of MIL-STD-810, Method 502.1, Procedure I, except the temperature shall be -48°C. Steps 4 and 5 shall be omitted. The motor samples shall then be operated in accordance with 4.3.3. To be acceptable, the motors shall meet the requirements of 3.2.1.1.

4.3.2.3 Vibration (random). The testing shall be conducted in accordance with MIL-STD-810. Time of testing shall be 120 minutes in

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each of the three major orthogonal axes at low temperatures of -48 ±3°C.

a. The vibration test profile shall be in accordance with Figure 3.

b. The tolerance on vibration test levels and test conduct shall be in accordance with MIL-STD-810, Method 514.2, paragraph 4.5.2, "Random Vibration Test."

c. Test fixture defined by Drawing SA2875556 shall be utilized.

The motor samples shall be operated in accordance with 4.3.3. To be acceptable, the motor shall meet the requirement of 3.2.1.1.

4.3.2.4 Mechanical shock. Testing shall be performed in accordance with MIL-STD-810, Method 516.2, Procedure I with the modifications and additions herein. Testing shall be conducted by applying three shocks in each direction through three mutually perpendicular principal axes at a temperature of -48 ±3°C, as follows:

a. The shock pulse signature shall be a half sine pulse, 15 g peak and 11 ms duration. The shock pulse tolerance limits shall be in accordance with MIL-STD-810, Method 516.2, Figure 516.2-2, "Half Sine Shock Pulse Configuration and its Tolerance Limits."

b. Before applying mechanical shocks, the motor samples shall be temperature conditioned at -48 ±3°C for not less than 4 hours. If the motor samples must be removed from the chamber to apply the required shocks, the elapsed time between removal from the chamber and the start of application of any shock along any axis shall not exceed 3 minutes. If the motor samples are transferred from the temperature conditioning chamber to a temperature conditioned insulated box which is at the same temperature, the elapsed time between removal from the chamber and start of application of any shock along any axis may be increased to 5 minutes, provided the motor

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samples are not exposed to ambient temperature for more than 3 minutes of the 5-minute period. If shock testing is not completed within the time limits specified, the motor samples shall be returned to the conditioning chamber for not less than 1 hour before shock tests are resumed.

c. The motor samples shall then be operated in accordance with 4.3.3. To be acceptable, the motor samples shall meet the requirements of 3.2.1.1.

4.3.2.5 Temperature shock. The motor samples shall be subjected to the temperature shock test of MIL-STD-810, Method 503.1, Procedure I, except the high temperature shall be +74°C and the low temperature shall be -48°C. To be acceptable, the motors shall meet the requirements of 3.2.1.1 when operated in accordance with 4.3.3.

4.3.3 Functional (cold gas). Unless otherwise specified in the contract or purchase order, testing shall be performed at ambient conditions. The test set-up shall conform to Figure 4. Testing shall be conducted as follows:

a. The hot gas motor shall be connected to the water brake through a 1 to 1 gear ratio.

b. The water break shall be adjusted to demand a 7.0 horsepower input to achieve a 2,200 ±100 rpm shaft speed.

c. Direct the nitrogen flow through the SA495191 orifice and adjust the pressure regulator so that the pressure downstream of the 0.206 diameter orifice is 1,100 +50, -0 psig. (This should result in proper flow rate.)

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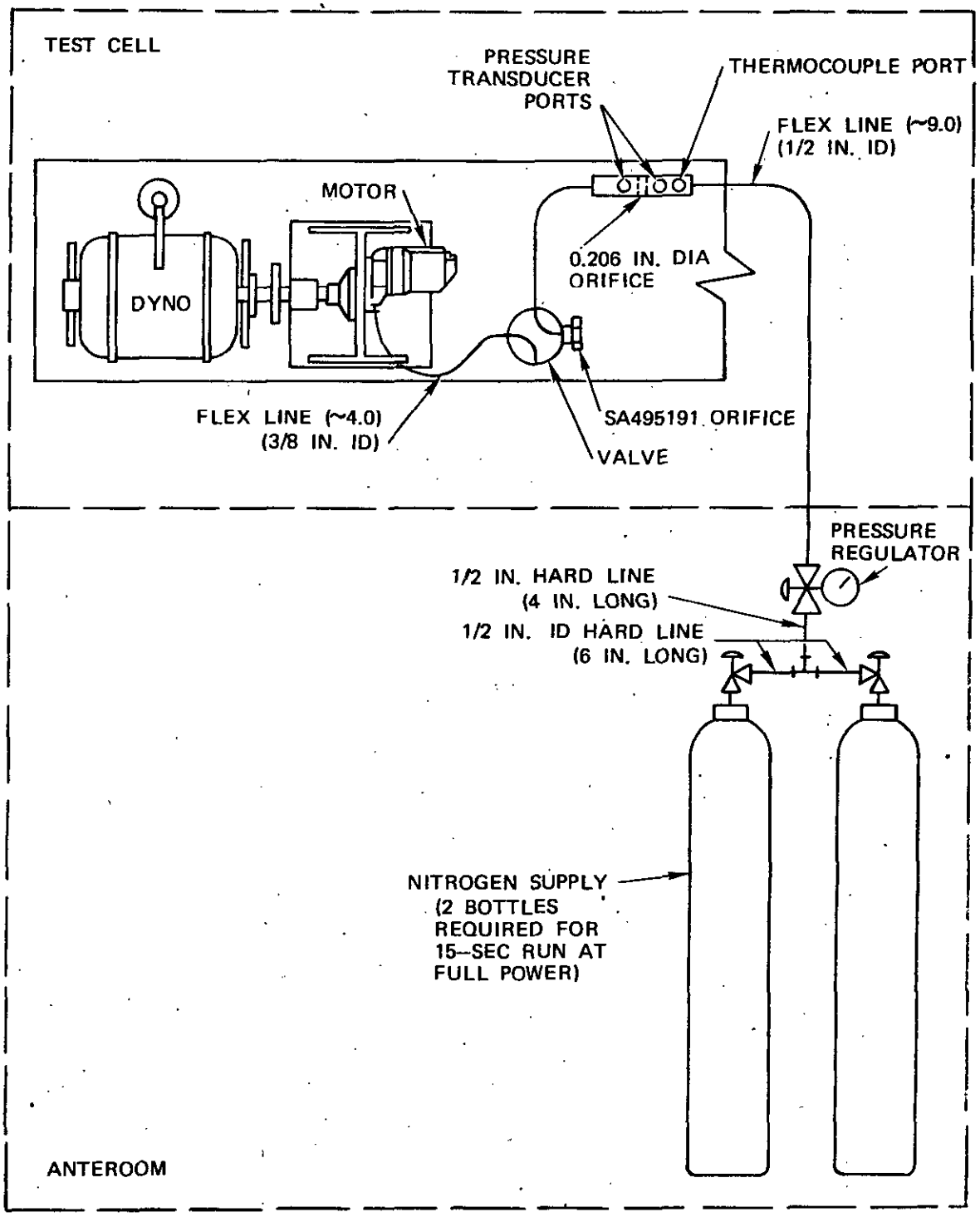


FIGURE 4. Bigeye motor cold gas test set-up.

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d. The valve shall then be repositioned, directing the gas flow to the motor. Allow the motor to run for 7.5 ±2.5 seconds. Motor shaft speed, motor torque, and gas flow rate shall be continuously maintained.

To be acceptable, the motor shall meet the requirements of 3.2.1.1.

4.3.4 Cold gas spin-up. Each motor shall be operated using nitrogen. Motor inlet pressure shall be regulated such that motor speed is between 2,000 and 2,500 rpm. Each motor shall be operated for 5 seconds minimum. Motors not meeting the requirements of 3.3.5 or showing evidence of improper assembly shall be rejected.

4.4 Test equipment and conditions.

4.4.1 Standard test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements at the rate of delivery specified in the contract or purchase order. The calibration of measuring and test equipment shall be in accordance with MIL-STD-45662. Test equipment, including associated calibration programs and operation and maintenance procedures, are subject to disapproval by the Government representative. When special test equipment and circuits are devised or commercially available equipment is employed, all test equipment circuits and methods are subject to disapproval by the Government representative. When special test equipment is employed, such equipment circuits and methods are subject to disapproval by the Government representative. Unless otherwise specified herein, the magnitude of any error introduced by test equipment shall not exceed 10 percent of the tolerance of the requirement being measured.

4.4.2 Test conditions. Unless otherwise specified herein, all tests shall be conducted at the test area ambient temperature and relative humidity. Where special conditions of tests are specified, equipment employed shall adequately provide specified conditions, and all monitoring and measurement devices shall conform to the requirements of 4.4.1. When special conditions of tests are not imposed, commercially available direct-measurement equipment may be employed provided they conform to the requirements of 4.4.1.

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4.5 Packaging, packing, and marking. Prior to shipment, examination shall be made to assure that packaging, packing, and marking conform to Section 5 of this specification.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C in accordance with FED-STD-102. The motor shall be packaged to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Packages shall comply with the Uniform Freight Classification Rules or regulations of other carriers, as applicable to the mode of transportation.

5.2 Packing. Packing shall be Level C in accordance with FED-STD-102. The motor, packaged in accordance with 5.1, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with the Code of Federal Regulations, 49 CFR 171-178.

5.3 Marking.

5.3.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.3.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The motor is intended to be used to drive the central injector of the BLU-80/B weapon.

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6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 15 motors as specified in 3.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests and approval of the first article.

c. Responsibility for inspection and inspection facility if different than 4.1.

d. Functional (cold gas) if different than 4.3.3.

e. Special marking (see 5.3.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited as follows:

Paragraph	Data Requirement	Applicable DID
3.3.2.2, 4.1.2.3, 4.2.3	Certificate of compliance	DI-E-2121
3.5, 4.1.2.3, 4.2.3	Inspection report	DI-T-2072

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(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Safety precautions. The loading, assembly, and handling of the motor involve hazardous operations and require explosive safety precautions as stated in DoD 4145.26M.

Preparing activity:
Navy - AS

(Project 1377-N676)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-85255(AS), MOTOR, HOT GAS (MAU-132A/B)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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